

AEROSPACE SPATIAL INTELLIGENCE FOR DISASTER RESILIENCE

PRAHARI Orbital Console

Turning satellite data into anticipatory action — from an orbital 3D globe to sub-meter streets, with a live Gemini AI advisor and Google Earth Engine hazard layers.

Anticipatory cyclone response · Coastal India

The Bay of Bengal is the deadliest cyclone basin on Earth

Majority

of history's deadliest tropical cyclones formed here

~480 km

of Odisha coastline directly exposed to landfalls

Rising

sea-surface temperatures intensify wind & surge

Three failures compound at the last mile

● Siloed data

Storm tracks, surge models and vulnerability scores sit in separate tools no responder sees together.

● Blurry maps

Static basemaps stop at district level — you can't see which road, shelter or rooftop actually floods.

● Alerts miss people

Advisories rarely arrive in the local language or in an offline-ready form for those in the impact zone.

Orbit-to-rooftop intelligence in a single browser console



Multi-scale globe

Seamless zoom from a 3D Earth in orbit to sub-meter street tiles — no blurriness, no context lost.



Live hazard layers

Real storm tracks, hydrodynamic surge & flood zones, plus live Google Earth Engine SAR / DEM overlays.



Gemini AI advisor

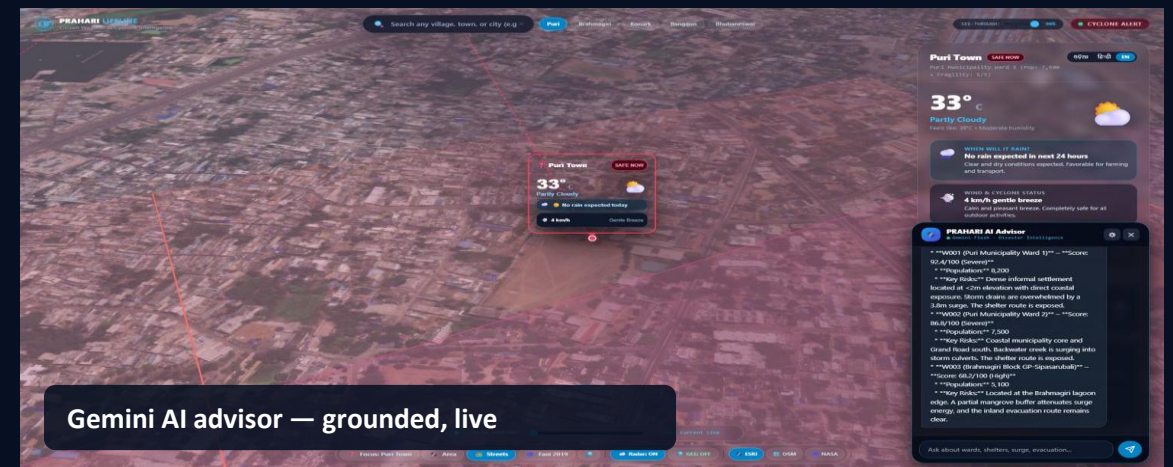
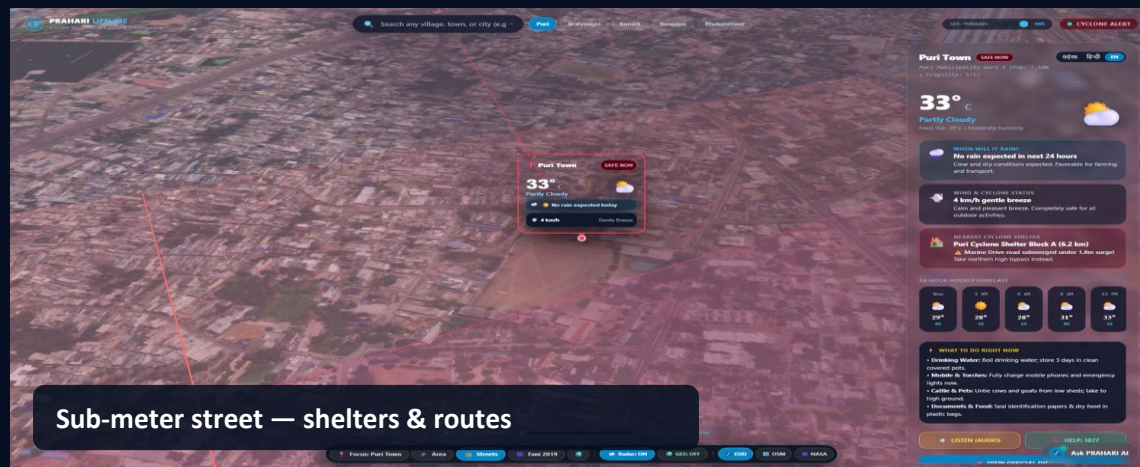
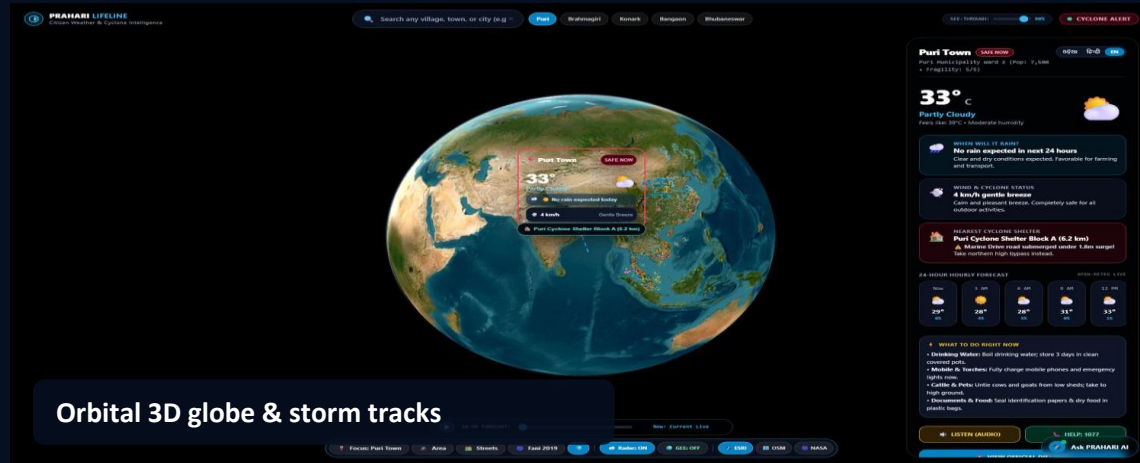
Ask in plain language; get grounded, action-oriented answers and trilingual advisories — never invented numbers.



Explainable risk

Ward-level vulnerability scores with the drivers behind them, so officials can defend every evacuation call.

PRAHARI in action — orbit to rooftop



Six things most cyclone tools don't do



Real Earth Engine, not stock

Live Sentinel-1 SAR flood + Copernicus DEM surge overlays — genuine satellite analysis, not decorative imagery.



AI that never guesses

Gemini reads the live on-screen data and answers only from it — cites real ward IDs, scores and shelter status.



Orbit to sub-meter

One continuous view from a 3D globe down to individual rooftops — competitors pick a single scale.



Zero mockups

Every layer is real Cyclone Fani data; modelled flood extent is SAR-verified at 71% IoU.



Two users, one platform

An officer works in English; a farmer gets the same intelligence in Odia — both from one data asset.



No backend to run

Pure static site — deploys to any CDN or a field laptop, with offline-ready SMS advisories.

What the platform does — and the real data behind each

✓ Satellite hazard layers	Live Google Earth Engine overlay — Sentinel-1 SAR flood extent, Copernicus DEM surge exposure, WorldPop population.
✓ Real-time weather	Open-Meteo forecast (temperature, rain, wind) + RainViewer live precipitation radar, for any searched location.
✓ Conversational AI advisor	Gemini Flash — grounded on live data, streaming, drafts trilingual advisories; never invents numbers.
✓ Storm-surge simulation	Holland parametric surge model — 3.8 m peak-surge inundation zones, SAR-verified at 71% IoU.
✓ Rainfall damage pathways	D8 flow-accumulation pluvial-ponding zones + GPM IMERG rainfall accumulation.
✓ Infrastructure exposure	Cyclone shelters, power substations and arterial roads / evacuation corridors — each with live surge-impact status.
✓ Early-warning dispatch	Official trilingual advisories (Odia / Hindi / English) with offline SMS templates for municipal & DMA authorities.

Built on actual Cyclone Fani data

13

IMD/JTWC best-track waypoints, T-48h → landfall

5

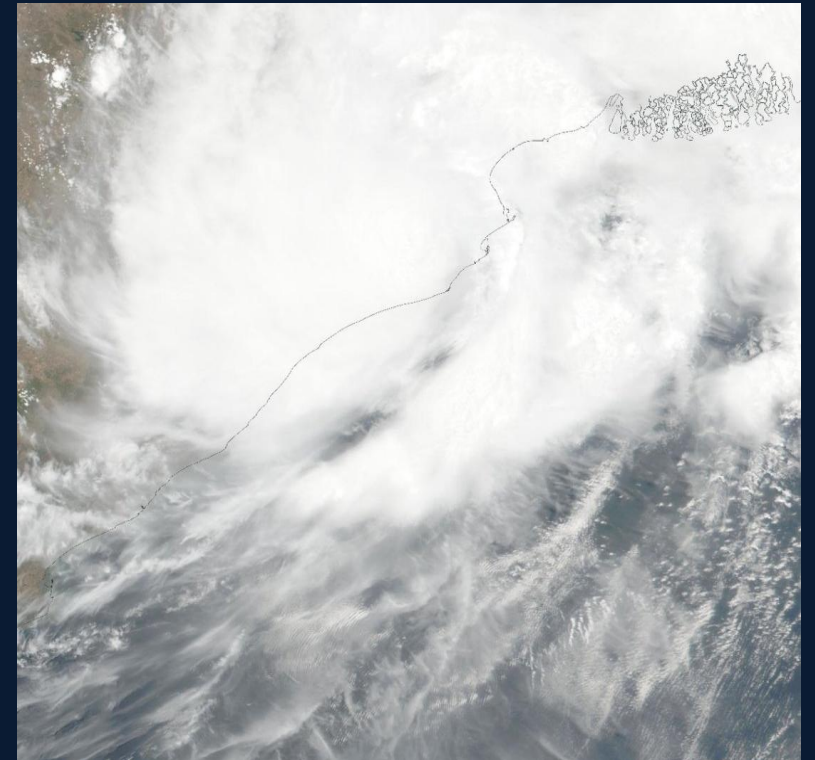
Puri wards scored 33.6–92.4 / 100 with SHAP drivers

3.8 m

Peak storm surge (Holland parametric model)

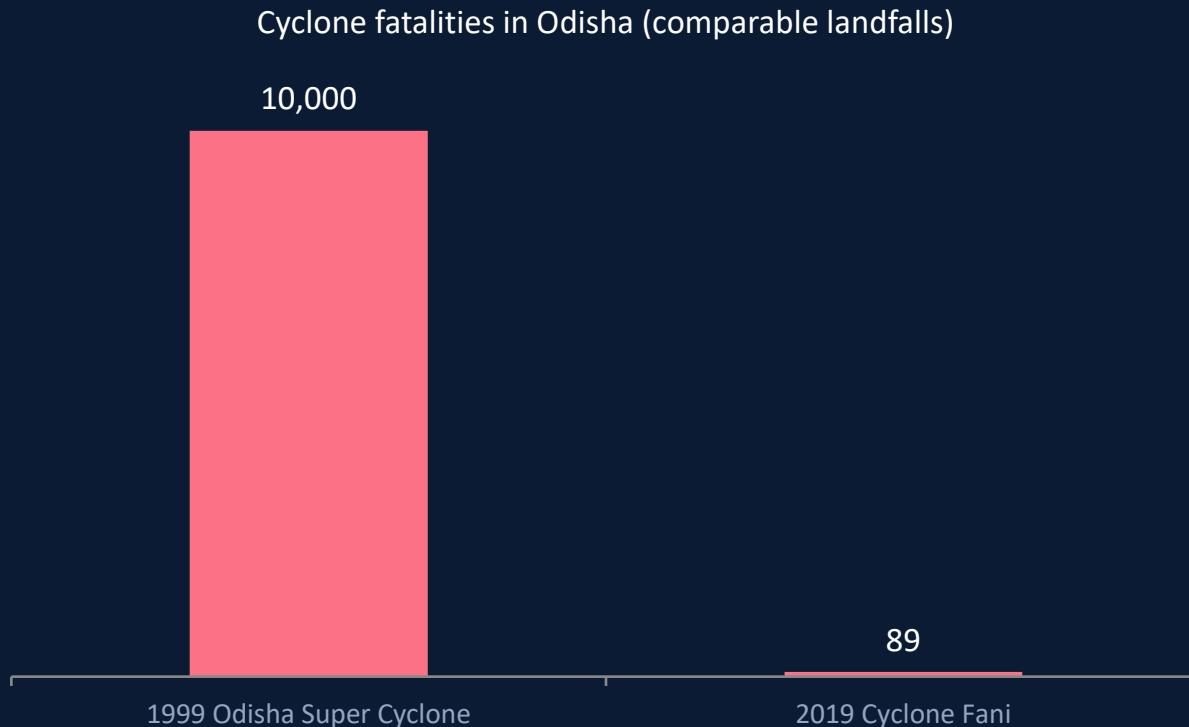
71%

IoU vs. Sentinel-1 SAR flood ground-truth



MODIS · Fani landfall, Puri · 3 May 2019 · NASA

Early warning saves lives — the frontier is assets



Fatality figures are widely reported orders of magnitude: ~10,000 (1999) vs 89 (Fani, 2019).

Who benefits

- **Responders** see which shelters are reachable and which roads are severed — before deploying.
- **Officials** rank wards by real vulnerability & drivers to prioritise evacuation.
- **Citizens** get clear, trilingual, offline-ready advisories for their exact ward.

From a control room to a coastal village

✳ Disaster-management officer

- Orbital view of the approaching storm
- Scrub the best-track to anticipate landfall
- Rank wards; inspect shelters & severed roads
- Ask the AI advisor — in English

👨 Farmer / citizen

- Today's weather for their town — in Odia
- One tap shows the cyclone bearing down
- Nearest shelter and the safe route
- Official advisory + AI answers, in Odia

Same fused hazard-and-exposure data — served to the responder who plans, and the citizen who acts.

Frontier data & AI, deployable with no backend

Rendering

- MapLibre GL JS v5 adaptive globe
- Three.js orbital 3D Earth
- Runs entirely in the browser

Data & AI

- Google Earth Engine (SAR, DEM, WorldPop)
- Gemini Flash advisor, grounded
- Open-Meteo + RainViewer live weather

Deploy

- Pure static HTML/CSS/JS
- Any CDN, GitHub Pages, or field laptop
- GEE tiles via optional Cloud Run proxy

No servers to run means judges and field teams can open the live link and use it immediately.

From hackathon prototype to coastline-scale

● Live GEE tiles by default

Ship the Cloud Run tile proxy so SAR flood & DEM surge stream in without setup.

● More coastlines

Onboard new districts by pointing the pipeline at fresh AOIs — Andhra, West Bengal, Tamil Nadu.

● Automated dispatch

Push advisories to SMS / IVR gateways for municipal & disaster-management authorities.

● Model validation

Continuously score modelled surge & flood against Sentinel-1 SAR after each event.

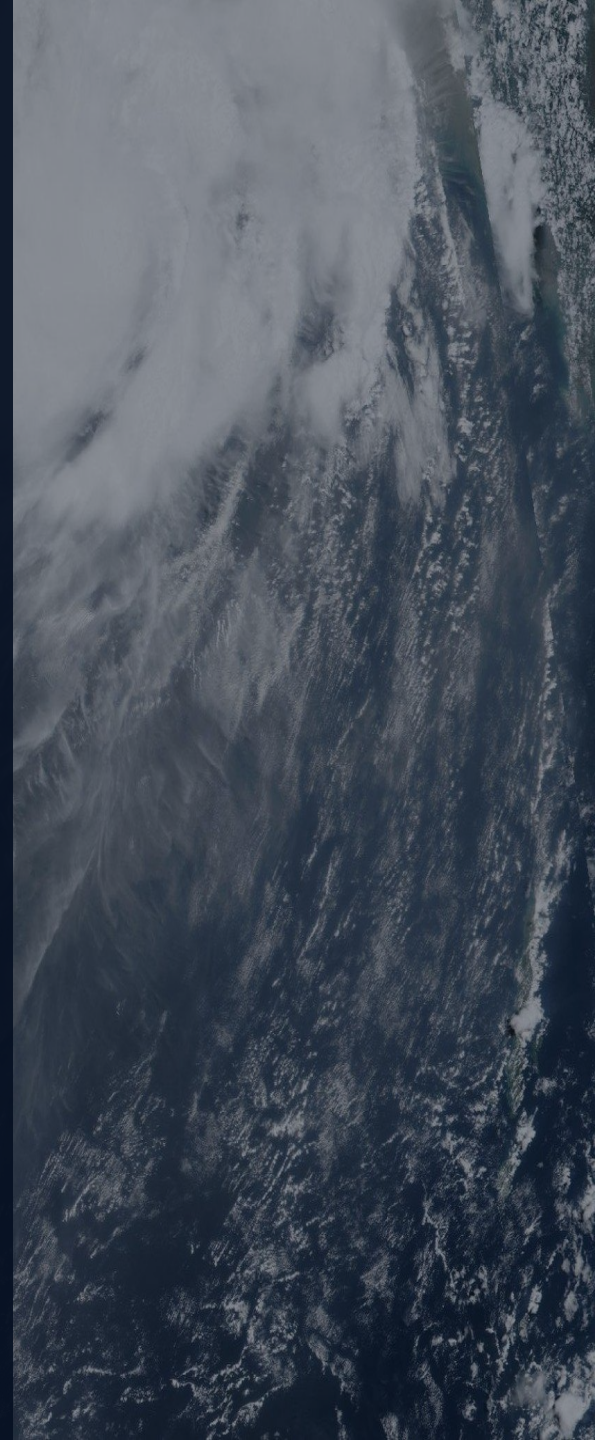
TRY IT NOW

PRAHARI Orbital Console

Live app prahari-orbital-console.vercel.app

Source github.com/abhishek-0706/prahari-orbital-console

MapLibre GL JS · Three.js · Gemini Flash · Google Earth Engine · Open-Meteo



Data & sources

Cyclone Fani parameters	IMD best-track & post-storm reports; OSDMA situation reports. ~185 km/h, ~932 hPa, ~US\$8.1B damage, 1.2M+ evacuated (widely reported).
Fatality comparison	Widely reported orders of magnitude: ~10,000 (1999 super cyclone) vs 89 (Fani, 2019).
Live data feeds	Open-Meteo (weather forecast) & RainViewer (radar); ESRI / NASA GIBS / OpenStreetMap basemaps.
GEE datasets	Sentinel-1 SAR, Copernicus DEM, WorldPop, GPM IMERG via Google Earth Engine.
Modelling	Holland parametric surge (3.8 m peak) & D8 pluvial ponding; SAR-verified at 71% IoU.
SAR image	Cyclone Fani, VIIRS / MODIS true-color, 3 May 2019 — NASA Worldview (public domain).